MATEMATICI 2

Seminar 6

Probl. $f : V \to V$ liniară ; V sp. vect. real

$f \circ f = f$

Să se arate că f diagonalizabilă.

$f \circ f = f$

S-a dem. că val. prop. sunt 0 sau 1 și

$V_{\lambda_1} = \text{Ker} f$ și $V_{\lambda_2} = \text{Im} f$

$\lambda_1 = 0 \; ; \; \lambda_2 = 1$

Pt. că $\lambda_1 \neq \lambda_2 \Rightarrow V_{\lambda_1} \oplus V_{\lambda_2} \Rightarrow \operatorname{Ker} f \oplus \operatorname{Im} f$

Pt. că $\dim \operatorname{Ker} f + \dim \operatorname{Im} f = \dim V \Rightarrow$

$\Rightarrow \dim (\operatorname{Ker} f \oplus \operatorname{Im} f) = \dim V$

$= \dim \operatorname{Ker} f + \dim \operatorname{Im} f$

$\Rightarrow \operatorname{Ker} f \oplus \operatorname{Im} f = V \Rightarrow V_{\lambda_1} \oplus V_{\lambda_2} = V \xRightarrow[\text{f. diag.}]{\text{Cf.}} f$ diag.

Formând matricea lui f, pe diag. vor fi numai 0 și 1.
$\quad\quad\quad

ex. Fie $A = \begin{pmatrix} 11 & -5 & 5 \\ -5 & 3 & -3 \\ 5 & -3 & 3 \end{pmatrix}$

1) Să se diagonalizeze A

2) Să se găsească B a.î. $B^2 = A$

3) Să se calculeze A^{2019}.

1) $\det(A - \lambda I_3) = 0$

$$\begin{vmatrix} 11-\lambda & -5 & 5 \\ -5 & 3-\lambda & -3 \\ 5 & -3 & 3-\lambda \end{vmatrix} = 0$$

$$= \begin{vmatrix} 11-\lambda & -5 & 0 \\ -5 & 3-\lambda & -\lambda \\ 5 & -3 & -\lambda \end{vmatrix} = \begin{vmatrix} 11-\lambda & -5 & 0 \\ -10 & 6-\lambda & 0 \\ 5 & -3 & -\lambda \end{vmatrix} =$$

$$= (-\lambda) \begin{vmatrix} 11-\lambda & -5 \\ -10 & 6-\lambda \end{vmatrix} = (-\lambda)(66 - 6\lambda - 11\lambda + \lambda^2 - 50) =$$

$$= (-\lambda)(16 - 17\lambda + \lambda^2) = 0$$

$= (-\lambda)(16 - 17\lambda + \lambda^2) = 0$

$\lambda_1 = 0 \qquad n_1 = 1$

$\Delta = 289 - 64 = 225 = 15^2$

$\lambda_{2,3} = \dfrac{17 \pm 15}{2}$

$\lambda_2 = \dfrac{32}{2} \Rightarrow \lambda_2 = 16 \qquad n_2 = 1$

$\lambda_3 = \dfrac{2}{2} \Rightarrow \lambda_3 = 1 \qquad n_3 = 1$

$$D = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 16 \end{pmatrix}$$

$\dim V_{\lambda_1} \geq 1$

$\dim V_{\lambda_2} \geq 1$

$\dim V_{\lambda_3} \geq 1$

$3 \leq \dim V_{\lambda_1} + \dim V_{\lambda_2} + \dim V_{\lambda_3} =$

$= \dim (V_{\lambda_1} \oplus V_{\lambda_2} \oplus V_{\lambda_3}) \leq 3$

$\Rightarrow \dim V_{\lambda_1} = \dim V_{\lambda_2} = \dim V_{\lambda_3} = 1$

Fie $v \in \mathbb{R}^3$; $v = (x_1, x_2, x_3)$

$$f(v) = f(x_1, x_2, x_3) = (-2x_1 + x_2 + x_3,\ x_1 - 2x_2 + x_3,\ x_1 + x_2 - 2x_3)$$

$$f(x_1, x_2, x_3) = (-2x_1 + x_2 + x_3) \cdot (1,0,0) + (x_1 - 2x_2 + x_3) \cdot (0,1,0) + (x_1 + x_2 - 2x_3) \cdot (0,0,1)$$

$\det. M_f^B$

$$f(e_1) = f(1,0,0) = (-2, 1, 1) = -2e_1 + e_2 + e_3$$

$$f(e_2) = f(0,1,0) = (1, -2, 1) = e_1 - 2e_2 + e_3$$

$$f(e_3) = f(0,0,1) = (1, 1, -2) = e_1 + e_2 - 2e_3$$

$$M_f^B = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} = A$$! coord. se pun pe coloane!

det. val. prop.

$$\det(A-\lambda I_3) = \det\left(\begin{pmatrix}-2 & 1 & 1\\ 1 & -2 & 1\\ 1 & 1 & -2\end{pmatrix} - \begin{pmatrix}\lambda & 0 & 0\\ 0 & \lambda & 0\\ 0 & 0 & \lambda\end{pmatrix}\right) =$$

$$= \begin{vmatrix}-2-\lambda & 1 & 1\\ 1 & -2-\lambda & 1\\ 1 & 1 & -2-\lambda\end{vmatrix} = (-2-\lambda)^3 + 1 + 1 + 2+\lambda + 2+\lambda + 2+\lambda$$

$$= -(8+\lambda)^3 + 2 + 6 + 3\lambda = -(8 + 12\lambda + 6\lambda^2 + \lambda^3) + 8 + 3\lambda$$

$$= -8 - 12\lambda - 6\lambda^2 - \lambda^3 + 8 + 3\lambda$$

$$= -\lambda^3 - 6\lambda^2 - 9\lambda = -\lambda(\lambda^2 + 6\lambda + 9) =$$

$$= -\lambda(\lambda+3)^2 = 0$$

$$\begin{cases} \lambda_1 = 0 & n_1 = 1 \\ \lambda_2 = -3 & n_2 = 2 \end{cases}$$

$$V_{\lambda_1} = \{ a(1,0,0) + b(0,1,0) + c(0,0,1) \mid a,b,c \in \mathbb{R}, (M_f^B - \lambda I_3)\begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \}$$

$$V_{\lambda_1} = \{ (a,b,c) \mid a,b,c \in \mathbb{R}, (M_f^B - \lambda_1 I_3)\begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \}$$

$$M_f^B - \lambda_1 I_3 = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} - \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_1 & 0 \\ 0 & 0 & \lambda_1 \end{pmatrix} = \begin{pmatrix} -2-\lambda_1 & 1 & 1 \\ 1 & -2-\lambda_1 & 1 \\ 1 & 1 & -2-\lambda_1 \end{pmatrix}$$

$$= M_f^B = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}$$

$$\begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} -2a+b+c \\ a-2b+c \\ a+b-2c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$a+b = 2c \Rightarrow c = \frac{a+b}{2} \Rightarrow \boxed{c = a = b}$$

$$-2a+b+\frac{a+b}{2} = 0 \quad / \cdot 2$$

$$-4a+2b+a+b = 0$$

$$-3a+3b = 0$$

$$-a+b = 0 \Rightarrow \boxed{a = b}$$

$V_{\lambda_2} = \{(a, a, a) \mid a \in \mathbb{R}\}$ $\{(1,1,1)\}$ bază pt. V_{λ_1}

$$M_f^B - \lambda_2 I_3 = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} - \begin{pmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} a+b+c \\ a+b+c \\ a+b+c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$a+b+c = 0 \Rightarrow c = -a-b$

$V_{\lambda_2} = \{(a, b, -(a+b)) \mid a, b \in \mathbb{R}\}$

$b = \alpha$
$c = \beta$ $\Rightarrow a = -b - c = -(\alpha + \beta)$

$\Rightarrow (x, y, z) = \alpha \underbrace{(-1, 1, 0)}_{V_1} + \beta \underbrace{(-1, 0, 1)}_{V_2} \Rightarrow$

$\Rightarrow V_{h_1} = \{ \alpha v_1 + \beta v_2 \mid \alpha, \beta \in \mathbb{R} \} \Rightarrow$

$\Rightarrow \{v_1, v_2\}$ sist. generatori

Din $\alpha v_1 + \beta v_2 = (0,0,0) \Rightarrow \begin{cases} \alpha = 0 \\ \beta = 0 \end{cases} \Rightarrow \{v_1, v_2\}$ liniar indep.

Ex.

Fie $f : V \to V$, V sp. vect. real finit, f liniară

$f \circ f = f$

Să se det. val. proprii și subsp de vect. proprii

Rez.

Obs. f sn. proiector

Fie $v \in V$, $v \neq 0$

„$\Rightarrow$" $f(f(v)) = f(v) = \lambda v =$

$= f(\lambda v) = \lambda f(v) = \lambda^2 v$

$\lambda^2 v - \lambda v = \lambda(\lambda - 1)v = 0 \Rightarrow$ $\lambda_1 = 0$, $\lambda_2 = 1$

Cum $v \neq 0$

$V_{\lambda_1} = \{ v \in V \mid f(v) = \lambda_1 v \} = \ker f$

$f(v) = 0 \cdot v = 0$

$f(f(v)) = f(0 \cdot v) = 0 \cdot f(v) = 0$

$f(0) = 0$

$V_{\lambda_2} = \{ v \in V \mid f(v) = \lambda_2 v \}$

$f(v) = 1 \cdot v = v$

$f(v) = v \Rightarrow v \in \mathrm{Im} f \Rightarrow V_{\lambda_2} \subseteq \mathrm{Im} f$ (1)

Fie $f(v) \in \mathrm{Im} f$

$f(f(v)) = 1 \cdot f(v) \in \mathrm{Im} f \Rightarrow$

$\Rightarrow f(v) \in V_{\lambda_2} = \{ w \in V \mid f(w) = 1 \cdot w \}$ $\Big| \Rightarrow$

$\Rightarrow \mathrm{Im} f \subseteq V_{\lambda_2}$ (2)

Din (1) și (2) $\Rightarrow \mathrm{Im} f = V_{\lambda_2}$